NATPOWER DIGITAL × CMA CGM · ALBERT CANAL

Trip report · Antwerp to Genk

Scenario 3 · Tender at the lock · Dry summer · CMA CGM e-barge, 135 m, two 2.9 MWh ZESpacks

88 km
route length, 6 locks

22 h 53
total time on the water, with group lockage

5.3 MWh
energy at the propeller

2.6 t
CO2 saved against a diesel barge

Battery charge along the route

Shown as the share of usable energy in the two packs. Without a swap the packs reach the 20% reserve at km 69 and run flat before the end of the trip. The swap at the waiting berth below Kwaadmechelen lock puts them back to 100%.



Both packs, with the swap · Same packs, no swap · 20% safe reserve

The day on one line

The tender meets the barge while it waits for the lock; the swap costs no time at all. In a dry summer the Meuse runs low and ships are locked through in groups, so each lock can take two to four hours.



Sailing 10 h 08 · Locks 12 h 00 · Waiting for the lock 15 min · Battery swap 30 min

Time and cargo space

What the swap costs in time, compared with charging the same two packs on board at a quay, and the cargo slots it gives back.



Time the stop adds to the trip

With swaps on the route: 2 packs on board

pack pack TEU TEU TEU

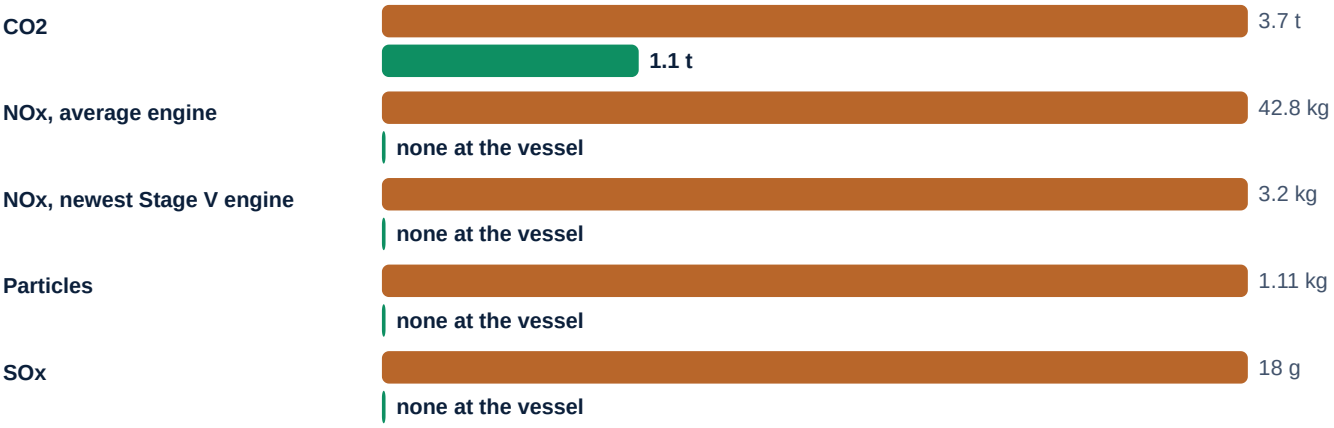
Round trip with no swaps: 5 packs on board

pack pack pack pack pack

Each 20 ft ZESpack takes one TEU slot. Five packs cover 176 km loaded with a 20% reserve, so carrying two gives three slots back to cargo.

Emissions on this trip

The same trip by a diesel barge and by the e-barge charged from today's Belgian grid. On renewable supply the e-barge CO2 falls close to zero, and it emits no NOx, particles or SOx at the quay or on the canal.



CO2 for the e-barge counts the electricity at the plug, including charging and drive losses. NOx, particles and SOx for the e-barge are zero at the vessel.

Diesel barge E-barge on swapped packs

Three ways to swap

Option	Time added	Equipment	Best when
Swap at the hub	30 min stop	Gantry crane and docking station	When the hub is on the route and has a free slot
Tender at the berth	None, inside the cargo call	Battery tender with its crane	When the barge already stops for cargo
Tender at the lock this report	None, inside the lock wait	Battery tender with its crane	When locks are busy; even more in dry summers

Assumptions and sources

Loaded e-barge about 60 kWh per km; each 2.9 MWh ZESpack has 2.6 MWh usable; 20% reserve; charging and drive losses 12%. Diesel: 0.22 kg fuel per kWh and 3.17 kg CO2 per kg; NOx 8.1 g per kWh for the average engine and 0.6 g for a Stage V engine (real world measurements on the Rhine, ACP 2023); particles 0.21 g and SOx 0.37 kg per TJ (German national inventory, inland navigation). Belgian grid 182 g CO2 per kWh in 2025 (Electricity Maps). Charging two packs at a quay takes about 3 hours from a 2 MVA connection. Dry summer: group lockage when the Meuse runs low, waits of 2 to 4 hours (Wijnegem and Olen, August 2022). Real figures will replace these once the operator, terminal and grid data are in.